

12 (**long**) Two carts of mass m , connected by a spring of stiffness k , are rolling without friction on a horizontal track, as shown in Fig. 9(a). Motion of cart 1 is described by its displacement x and that of cart 2 by its displacement y .

- (a) (i) Write down (or derive) a matrix equation of motion for the system of carts. [5]
 (ii) By inspection or otherwise find the natural frequencies and mode shapes. [5]

(b) A second spring, also of stiffness k , is used to anchor cart 1 to the ground, as shown in Fig. 9(b). A sinusoidal force f is applied to cart 1 at frequency ω such that $f(t) = F \cos(\omega t)$ and the responses are $x(t) = X \cos(\omega t)$ and $y(t) = Y \cos(\omega t)$. The amplitude of motion of cart 1 is given by the expression

$$X(\omega) = \frac{A(1 - \omega^2/\omega_0^2)}{(1 - \omega^2/\omega_1^2)(1 - \omega^2/\omega_2^2)}$$

- (i) By solving the equations of motion or otherwise find expressions for A , ω_0 , ω_1 and ω_2 in terms of k , m and F . [15]
 (ii) Sketch the variation of $X(\omega)$ with angular frequency ω . Indicate on your sketch the values of ω_0 , ω_1 and ω_2 . [5]

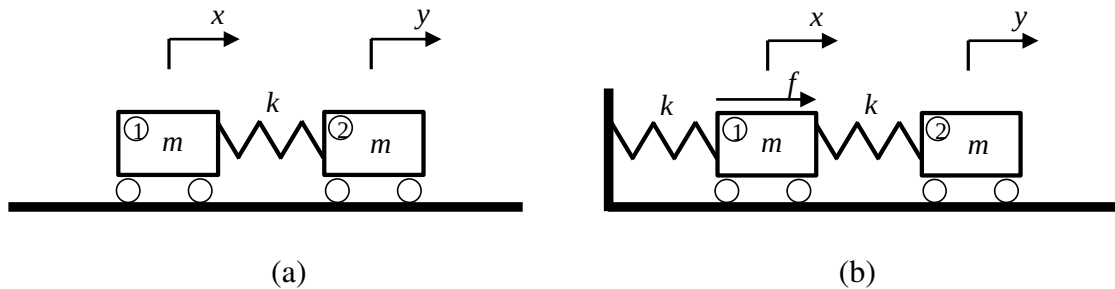


Fig. 9

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